JEE ADVANCED MOCK TEST - 1

PAPER 1: MATHEMATICS

SECTION - I (Single Correct Choice Type)
This section contains **5 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** choice is correct.
Marking Scheme: +4 for a correct answer, -1 for an incorrect answer, 0 if unattempted.

1. If $\tan\alpha = \frac{1}{7}$ and $\sin\beta = \frac{1}{\sqrt{10}}$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\alpha + 2\beta$ equals:

(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{3}$
(C) $\frac{3\pi}{4}$
(D) $\frac{\pi}{2}$

2. If $a = \tan 15^\circ$, $b = \csc 75^\circ$ and $c = 4\sin 18^\circ$, then:

(A) $c > b > a$
(B) $a > c > b$
(C) $b > a > c$
(D) $b > c > a$

3. The value of $\sin\frac{2\pi}{7}\sin\frac{4\pi}{7} + \sin\frac{4\pi}{7}\sin\frac{8\pi}{7} + \sin\frac{8\pi}{7}\sin\frac{2\pi}{7}$ equals:

(A) 0
(B) $\frac{1}{2}$
(C) $-\frac{1}{2}$
(D) 1

4. The number of real solutions of the equation $x^{2\log_x(x+3)} = 16$ is/are:

(A) 0
(B) 1
(C) 2
(D) Infinite

5. If $\log_{(24\sin\theta)}(24\cos\theta) = \frac{3}{2}$, where $\theta \in \left(0, \frac{\pi}{2}\right)$, then $\cot\theta$ equals:

(A) 2
(B) 4
(C) 8
(D) 16

SECTION - II (Multi-Select Choice Type)
This section contains **8 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE** choice is correct.
Marking Scheme:
- **Full Marks (+4):** If only the correct options are chosen.
- **Partial Marks (+3):** If all four options are correct, but only three are chosen.
- **Partial Marks (+2):** If three or more options are correct, but only two correct ones are chosen.
- **Partial Marks (+1):** If two or more options are correct, but only one correct option is chosen.
- **Zero Marks (0):** If the question is left unattempted.
- **Negative Marks (-2):** If any incorrect option is selected.

6. If $a = 2^{\log_2(\cos x)}$ and $b = e^{\ln(\tan x)}$, then the product (ab) cannot be equal to:

(A) $-\frac{\sqrt{2}}{\sqrt{3}}$
(B) 0.9
(C) 1
(D) $\frac{\sqrt{3}}{\sqrt{5}}$

7. Let $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + \cdots + \sin n\theta \sin n^2\theta)$, then which of the following hold(s) good?

(A) $f_3\left(\frac{\pi}{6}\right) = 0$
(B) $f_4\left(\frac{\pi}{3}\right) = 2$
(C) $f_8\left(\frac{\pi}{2}\right) = 0$
(D) $f_{11}\left(\frac{\pi}{4}\right) = 2$

8. Which of the following real number(s) is (are) non-positive?

(A) $\log_{0.\overline{3}}\left(\frac{\sqrt{5}+2}{\sqrt{5}-2}\right)$
(B) $\log_7(\sqrt{83} - 9)$
(C) $\log_{\tan\frac{\pi}{12}}\left(\cot\frac{\pi}{8}\right)$
(D) $\log_2\sqrt{9 \cdot \sqrt[3]{27^{-\frac{5}{3}} \cdot 243^{-\frac{7}{5}}}}$

9. If $x^{\frac{3}{4}(\log_3 x)^2 + \log_3 x - \frac{5}{4}} = \sqrt{3}$, then x has:

(A) one integral value
(B) two irrational values
(C) one irrational value
(D) two rational values

10. If $\sec\theta = \frac{5}{4}$ $\left(\frac{3\pi}{2} < \theta < 2\pi\right)$, then the value of the expression $\frac{7-5\cot\theta}{9-4\sqrt{\sec^2\theta-1}}$ is $\frac{a}{b}$ where a and b are coprime numbers. Then:

(A) $a + b = 59$
(B) $a - b = 23$
(C) $a + b = 29$

(D) $a - b = 17$

11. If the minimum and maximum values of $f(x) = 4\sin^4 x + 12\cos^2 x + 1$ are m and M respectively, then:

(A) $m + M$ is the square of a natural number
(B) $M - m$ is the cube of a natural number
(C) $2M - m$ is a prime number
(D) The sum of the digits of $(M + m)$ is 9

12. If $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} = 6$ and $\frac{a}{c} + \frac{b}{d} + \frac{c}{a} + \frac{d}{b} = 8$, then $\frac{a}{b} + \frac{c}{d}$ can be equal to:

(A) 4
(B) 2
(C) $\frac{1}{4}$
(D) $\frac{1}{2}$

13. Given $b^2 \geq 4ac$ for the equation $ax^4 + bx^2 + c = 0$. Then all the roots of the equation will be real if:

(A) $a > 0$, $b < 0$, $c > 0$
(B) $a < 0$, $b > 0$, $c > 0$
(C) $a < 0$, $b > 0$, $c < 0$
(D) $a > 0$, $b > 0$, $c > 0$

SECTION - III (Numerical Value / Integer Type)
This section contains **5 Numerical Value Type Questions**. The answer to each question is a single integer value.
Marking Scheme: +4 for a correct answer, -1 for an incorrect answer, 0 if unattempted.

14. If A, B and C are positive angles such that $A + B + C = \pi$, find the minimum value of the expression:

$$\frac{\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)} + \frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} + \frac{\cos\left(\frac{C-A}{2}\right)}{\cos\left(\frac{C+A}{2}\right)}$$

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15. If P denotes the product of all values of x satisfying the equation $\log_{3x}\left(\frac{3}{x}\right) + (\log_3 x)^2 = 1$, then find the value of $18P$.

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16. If $\sin\alpha + \sin\beta + \sin\gamma = 3$, find the value of $|\cos 2\alpha + \cos 2\beta + \cos 2\gamma|$.

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17. If M and m denote the maximum and minimum values of the expression $\sqrt{49\cos^2\theta + \sin^2\theta} + \sqrt{49\sin^2\theta + \cos^2\theta}$, then find the value of $(M - m)$.

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18. Positive integers a and b satisfy the condition $\log_2\left(\log_{2^a}\left(\log_{2^b}(2^{100})\right)\right) = 0$. Find the sum of all possible values of a.

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